PS 1: AI OS Applications that can bring positive change on AI OS

What is AI OS?

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An AI Operating System is a next-gen OS that has AI deeply embedded into its
core think of it like an OS that can understand context, automate tasks, and
make smart decisions natively (e.g., Apple Intelligence, Microsoft Copilot+
PC, or Android AI features).
What participants need to build:
An application that runs ON or integrates WITH an AI OS and creates a
meaningful, positive real-world impact  could be in healthcare, education,
accessibility, productivity, environment, etc.

It's open-ended and creative teams can pick any domain
The focus is on impact + innovation, not just technical complexity.

Possible skills use: App/Web development, AI/ML integration, UI/UX, API usage,
prompt engineering
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PS 2: Inference of AI Models in Non-GPU Environment

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What's the core problem?
Most AI/ML models (like LLMs, image classifiers) are trained and run on
powerful GPUs. But the real world doesn't always have GPUs think low-cost
laptops, Raspberry Pi, IoT devices, or rural servers. The challenge is to make
AI models run efficiently without a GPU.
What participants need to build:
A solution that can run an AI model on CPU-only hardware with acceptable speed
and accuracy. This involves model compression, optimization, and smart
engineering.
Why it's hard:

GPUs have thousands of cores for parallel processing; CPUs don't.
Large models are simply too slow/heavy on CPUs without optimization
Real-world deployment often happens on edge devices with zero GPU

Possible skills use: Model quantization, pruning, ONNX Runtime, llama.cpp,
TensorFlow Lite, OpenVINO, edge AI
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PS 3: Implement a program to detect duplicate records based on their name or description in a data set which contains records in different languages (for example: English, Japanese, Chinese, Thai, Bahasa, Arabic etc.)

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Here's a breakdown:
What's the core problem?
You have a large database with records (think product listings, user profiles,
entries, etc.) where the same item may appear multiple times  but written in
different languages or scripts. The goal is to build a system that can
identify these duplicates automatically.

For Example:
"Apple" in English and "りんご" in Japanese could refer to the same product
Spelling variations, transliterations, and cultural naming differences make
simple string matching useless
You need NLP/ML techniques like multilingual embeddings, fuzzy matching, or
semantic similarity

Key skills needed: NLP, multilingual models (e.g., mBERT, LaBSE), fuzzy string
matching, data deduplication, Python/ML frameworks
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